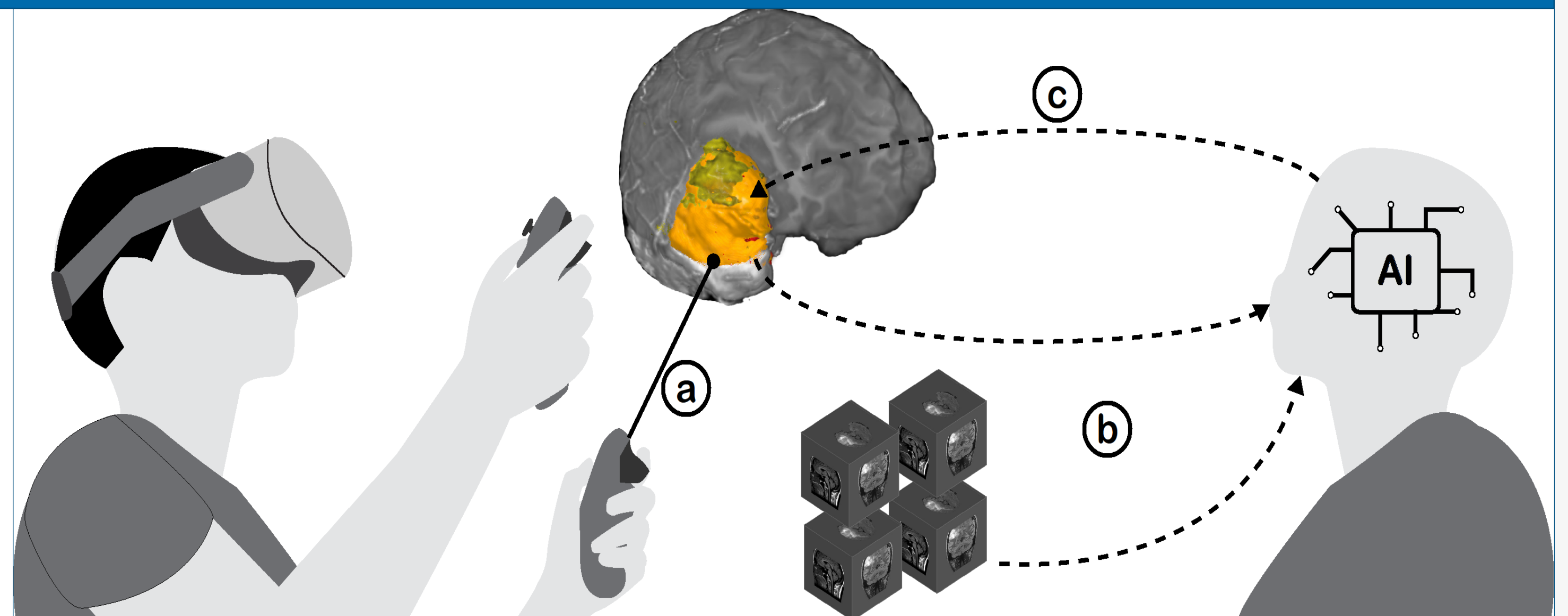


Introduction

Traditional tumour segmentation relies on manual annotation of three-dimensional scans, with experts marking hundreds of slices. The diverse appearance and unclear boundaries of brain tumours further complicate this process, which can delay diagnosis and limit treatment accessibility. Brain tumours differ significantly in shape, size, location, and contrast imperfections. Reliable segmentation is crucial for accurate identification and effective

treatment. Recent advances in AI-based 3D medical image segmentation, such as SAM-Med3D, have improved automation; however, expert supervision remains vital for brain imaging.

We present a virtual reality probe that integrates an expert-in-the-loop approach with SAM-Med3D, allowing users to iteratively enhance brain tumour segmentation by selecting points for AI refinement.



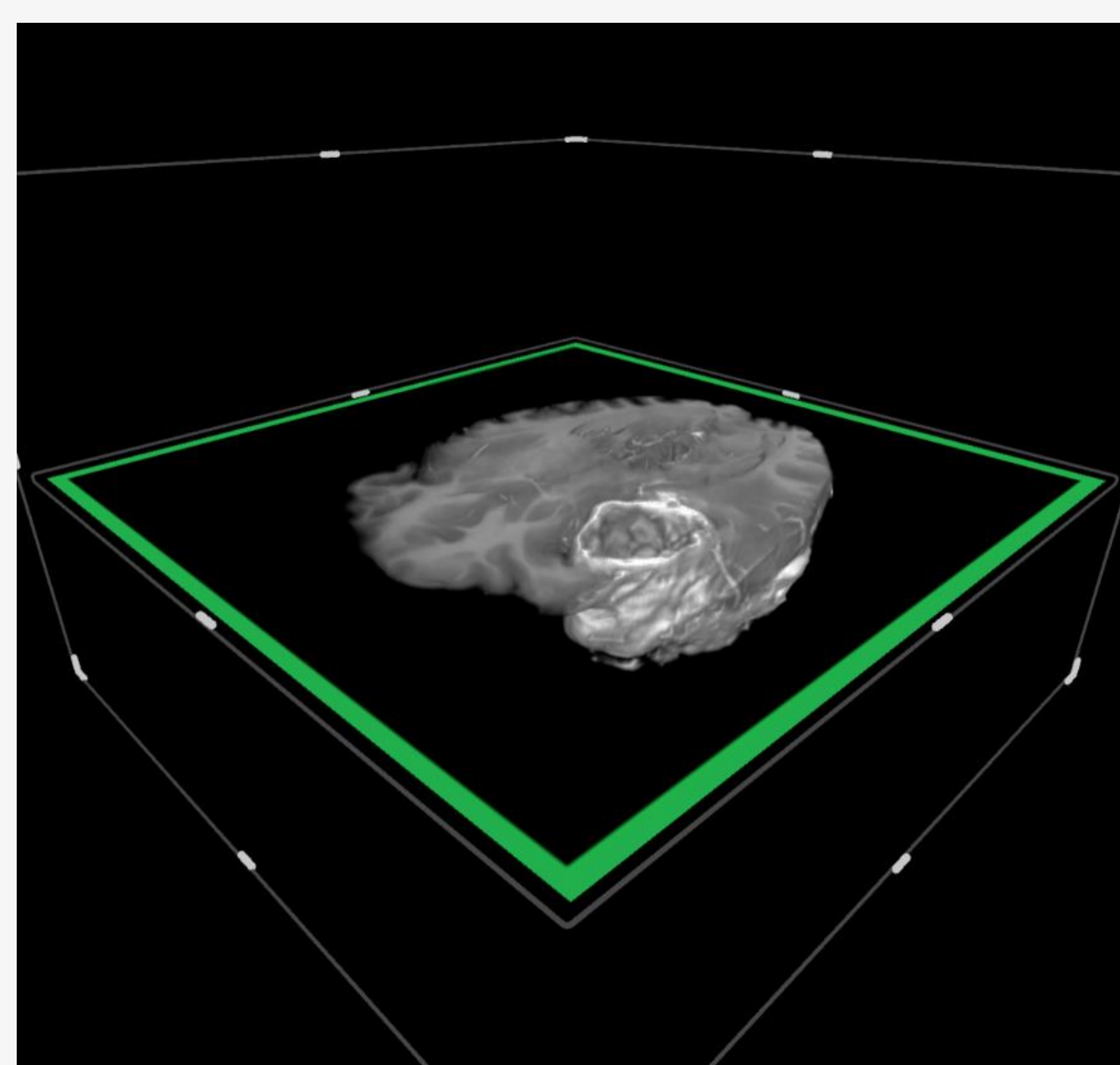
System Features

AI Model:

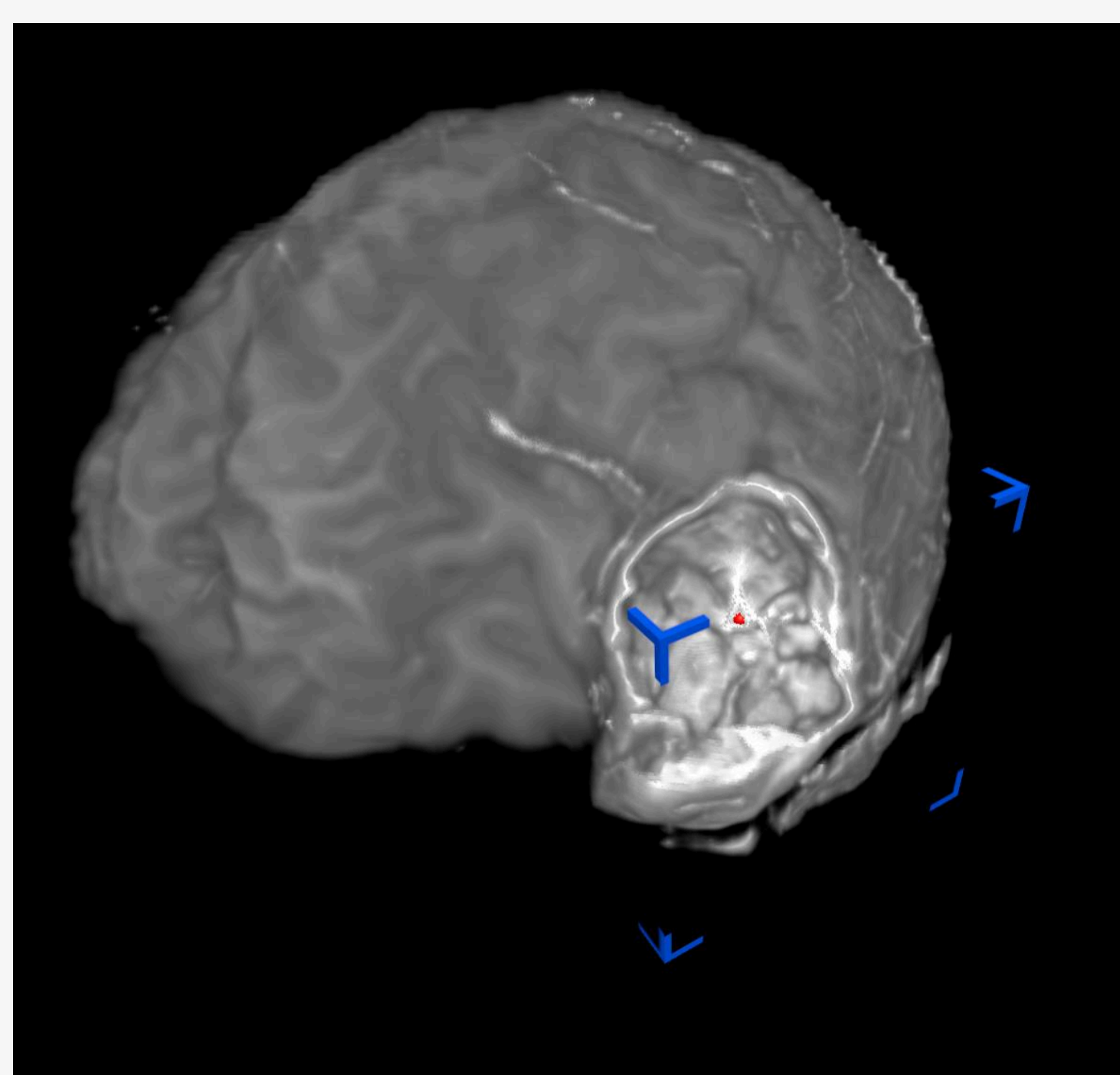
- Distinguishes three tumour subregions: Non-enhancing tumour (NCR/NET), Enhancing tumour (ET) and Edema (ED)
- Trained on 484 patient cases from MSD BraTS Task
- Incorporated a discriminator network to improve the segmentation quality
- Employed a SWIN Transformer to generate voxel embeddings
- Accepts MRI data, user points and previous masks (segmentation) as inputs

Visualisation:

- Basic natural navigation including zooming and rotation
- **Cross-section plane** is used to cut through the brain and segmented regions. User can freely move and rotate it around
- **Cutting box** is used to trim the surrounding regions
- **Point Selector** is used to make precise selection within a brain
- **Imaging Viewers** overlay the visualised brain with 2D planes to display actual imaging data for convenient analysis

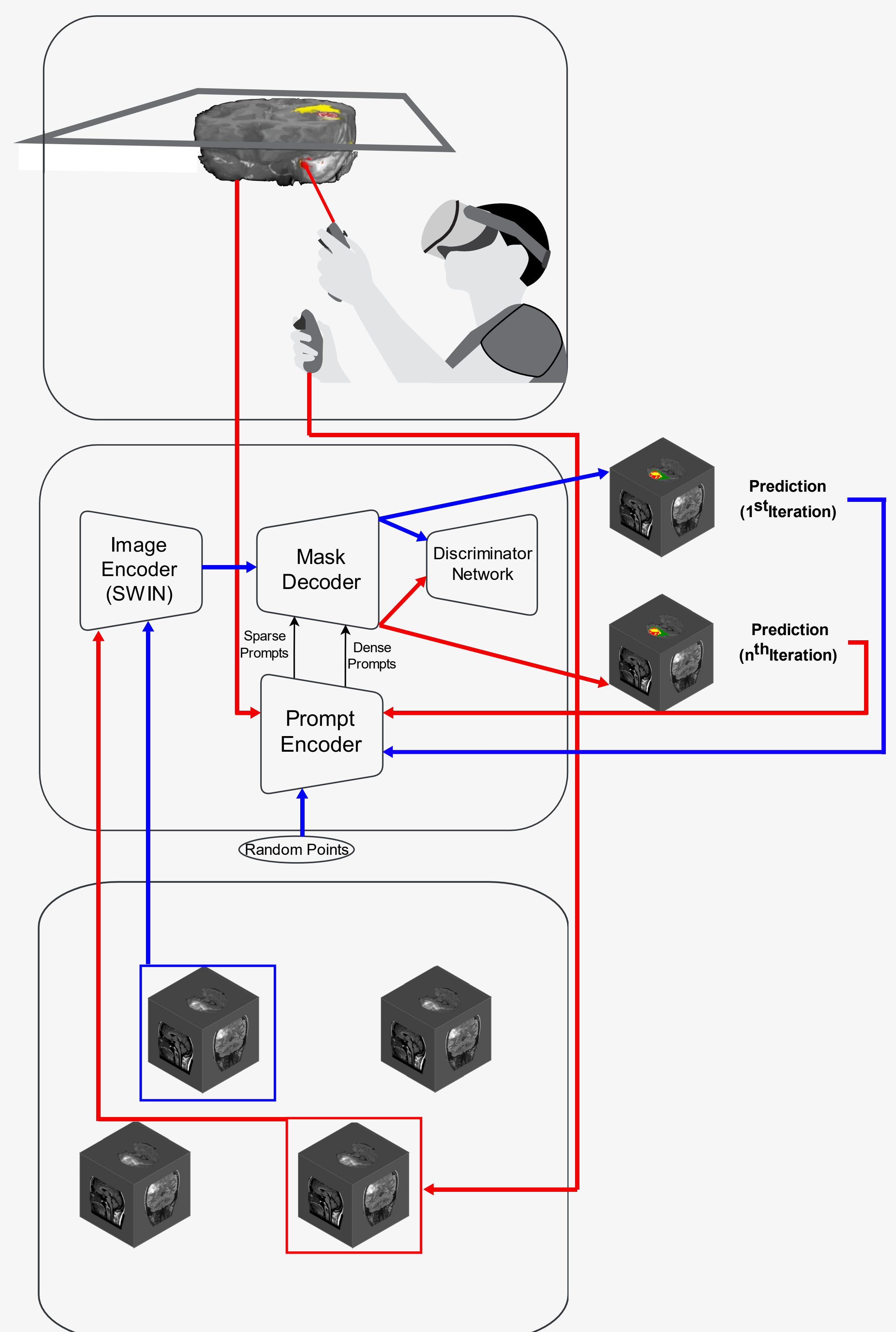


Cross-section Plane



Cutting Box and Point Selector (the red dot in the figure)

System Architecture



Discussion and Future Directions

Our proposed system is designed to provide experts with as much relevant and effective information as possible, enabling accurate and rapid decision-making.

By incorporating VR, we strive to achieve superior performance through quantitative comparison to traditional 2D desktop interfaces.

We believe that incorporating a human-in-the-loop approach is essential in the era of AI integration, particularly for sensitive decision-making tasks such as brain tumour diagnosis.

Immersive technologies offer a valuable platform for these tasks, but effective interaction and visualisation techniques are crucial to ensure informed decision-making while not hindering AI performance benefits.

In the future, we will test the effectiveness of this tool against automated and semi-automated desktop slicers. Additionally, future research may investigate advanced VR interaction techniques to enhance user performance and accuracy.



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